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This system uses **Generalized Autoregressive Conditional Heteroskedasticity (GARCH)** models to forecast stock price volatility and identify mispriced options. The core strategy is a **long straddle** · simultaneously buying an at-the-money (ATM) call and put option · on stocks where the GARCH model predicts that future realized volatility will exceed the market's current implied volatility.

- **Model:** GJR-GARCH(1,1,1) with Student-t innovations
- **Strategy:** Long straddle, 10 trading day hold
- **Budget:** \$150 per trade
- **Universe:** 42 affordable tickers (stocks priced ~\$1-\$30)
- **Execution:** Webull OpenAPI with automated order placement
- **Risk control:** Auto-close at +10% return; 10-day maximum hold

```

.                                     DATA LAYER
.   Yahoo Finance API · OHLCV, VIX, Treasury, Options
.   Local CSV cache (12-hour freshness)
.
.
.                                     MODEL LAYER
.   GARCH(1,1) vs GJR-GARCH(1,1,1)
.   Automatic AIC-based model selection
.   Student-t distribution for fat tails

```

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.....
.                               SIGNAL LAYER                               .
.   GARCH Realized Vol vs Market Implied Vol                             .
.   Spread > threshold · BUY VOL signal                                   .
.   Scanner: 42 tickers × 3 strikes each                                  .
.....

.....
.                               EXECUTION LAYER                           .
.   Webull OpenAPI (HMAC-SHA1 signed)                                     .
.   Two atomic SINGLE-leg orders (call + put)                             .
.   Auto-close monitor (30s polling)                                       .
.....

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Source files:

- `models/garch_model.py` · GARCH fitting, forecasting, diagnostics
 - `signals/scanner.py` · Universe scanning and signal ranking
 - `signals/generator.py` · IV vs RV spread computation
 - `backtest/rigorous.py` · Train/val/test evaluation
 - `broker/webull_client.py` · Order execution
 - `broker/auto_close.py` · Autonomous position monitoring
 - `data/fetcher.py` · Market data retrieval
 - `config.py` · All hyperparameters
-

3. The GARCH Model

3.1 What is GARCH?

GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models the fact that financial markets exhibit **volatility clustering** · periods of high volatility tend to follow high volatility, and calm periods follow calm periods.

Unlike simple historical volatility (which treats all past days equally), GARCH models learn how today's volatility depends on:

- 1 **Yesterday's volatility** (persistence)
- 2 **Yesterday's shock** (reaction to new information)

3.2 Model Specifications

Two models are fitted and compared:

Symmetric GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where:

- σ_t^2 = conditional variance at time t
- ω = long-run variance constant (omega)
- α_1 = ARCH coefficient · how much yesterday's squared shock affects today's variance
- β_1 = GARCH coefficient · how much yesterday's variance persists

- ϵ_{t-1} = yesterday's return shock (residual)

Constraint: $\alpha + \beta < 1$ ensures stationarity (variance doesn't explode).

Asymmetric GJR-GARCH(1,1,1)

$$\sigma_t^2 = \omega + (\alpha + \beta \cdot I_{t-1}) \cdot \epsilon_{t-1}^2 + \gamma \cdot \epsilon_{t-1}^2 \cdot I_{t-1}$$

Where:

- γ = leverage parameter · additional variance from negative shocks
- I_{t-1} = indicator function: 1 if $\epsilon_{t-1} < 0$, else 0

This captures the **leverage effect**: stock prices tend to become more volatile after negative returns than after positive returns of the same magnitude.

3.3 Innovation Distribution

Returns are modeled with a **Student-t distribution** rather than Gaussian, because financial returns exhibit:

- **Fat tails** · extreme moves occur more frequently than a normal distribution predicts
- **Excess kurtosis** · the distribution has heavier tails

The degrees-of-freedom parameter (ν) is estimated from data. Lower ν · fatter tails.

3.4 Model Selection

Both GARCH(1,1) and GJR-GARCH(1,1,1) are fitted, and the model with the **lower Akaike Information Criterion (AIC)** is selected:

$$AIC = 2k - 2 \cdot \ln(L)$$

Where k = number of parameters, L = likelihood. AIC penalizes model complexity · GJR-GARCH wins only if the leverage term significantly improves fit.

3.5 Configuration

Parameter	Value	Rationale	p (GARCH lag)	1	Standard; higher orders rarely improve fit
q (ARCH lag)	1	Captures immediate shock reaction	o (Leverage lag)	1	One asymmetric term for leverage effect
Distribution	Student-t	Fat tails in equity returns	Return scaling	×100	Numerical stability for optimizer
Trading days	252	Standard US market annualization			

4. Data Sources & Preparation

4.1 Price Data

- **Source:** Yahoo Finance API (via `yfinance` Python library)
- **Fields:** Open, High, Low, Close, Adjusted Close, Volume
- **Frequency:** Daily (close-to-close)
- **Date range:** January 1, 2022 · present (~1,081 trading days)
- **Caching:** CSV files in `cache/` directory, 12-hour freshness

4.2 Returns Computation

Log returns are used for GARCH fitting:

$$r_t = \ln(\text{Close}_t / \text{Close}_{\{t-1\}}) \times 100$$

The $\times 100$ scaling is applied for numerical stability during maximum likelihood estimation. All outputs are de-scaled before use.

4.3 Supplementary Data

| Data | Source | Purpose | |-----|-----|-----| | VIX (^VIX) | Yahoo Finance | Market fear gauge, regime detection
| | 10Y Treasury (^TNX) | Yahoo Finance | Risk-free rate for Black-Scholes pricing | | Options chains | Yahoo Finance (live) | Current implied volatility, strike prices, premiums |

4.4 Ticker Universe

42 tickers selected for affordability (stock price \$1-\$30, enabling ATM straddles within \$150):

F, SOFI, NIO, RIVN, SNAP, MARA, PLUG, LCID, AMC, GME,
BB, CLOV, OPEN, RIOT, SNDL, GEVO, ACHR, RGTI, QUBT, FCEL,
CHPT, QS, ENVX, RUN, XPEV, UPST, SKLZ, FUBO, CLSK, HIMS,
AAL, NCLH, PATH, BLNK, T, INTC, PFE, CCL, PYPL, DKNG,
HOOD, SIRI

5. Training & Model Selection

5.1 Data Splitting

The system uses a strict **chronological 60/20/20 split** to prevent look-ahead bias:

... Train (60%) ...	... Val (20%) ...	... Test (20%) ...
Jan 2022	Mid 2024	Late 2025
~648 days	~216 days	~217 days

- **Training set:** Used to estimate GARCH parameters ($\cdot, \cdot, \cdot, \cdot, \cdot$)
- **Validation set:** Used to verify GARCH forecast correlates with realized vol
- **Test set:** Completely unseen; used for backtesting trading performance

5.2 Fitting Procedure

- 1 Compute log returns from the training set price data
- 2 Scale returns by $\times 100$ for numerical stability
- 3 Fit GARCH(1,1) via Maximum Likelihood Estimation (MLE)
- 4 Fit GJR-GARCH(1,1,1) via MLE
- 5 Select the model with lower AIC
- 6 Extract conditional volatility series
- 7 De-scale and annualize: $\cdot_{\text{annual}} = (\cdot_{\text{daily}} / 100) \times \cdot_{252}$

The arch Python library (by Kevin Sheppard) handles MLE optimization using the SLSQP algorithm with automatic parameter bounds.

5.3 Diagnostics

After fitting, the system runs:

- **Ljung-Box test** on squared standardized residuals (lags 10, 20) · tests whether GARCH has captured all volatility clustering
- **Jarque-Bera test** on residuals · tests normality of standardized residuals
- **Parameter significance** · p-values for all estimated parameters

5.4 Validation Metric

On the validation set, the system computes the **Pearson correlation** between:

- GARCH conditional volatility forecast
- 5-day rolling realized volatility (from actual returns)

A correlation > 0.5 indicates the model successfully captures volatility dynamics out-of-sample.

6. Testing & Validation

6.1 Backtest Methodology

The backtest simulates the exact trading strategy on the **test set** (final 20% of data, never seen during training):

Entry rules:

- 1 Compute GARCH conditional vol at time t
- 2 Compute 21-day rolling market IV from historical returns
- 3 Calculate spread = GARCH_vol - Market_IV
- 4 If spread $> 2\%$ (entry threshold) · open a long straddle

Position sizing:

- Strike = round(spot price) · ATM
- Straddle price computed via Black-Scholes
- Max contracts = floor((capital $\times$ 90% - commissions) / straddle_cost)
- Commission: \$0.65 per contract per leg

Exit rules:

- Hold for exactly 10 trading days
- Exit value = intrinsic value of call + put at exit spot price
- No early exit, no stop loss (hold strategy outperformed stop-loss in testing)

6.2 Performance Metrics

The backtester computes:

Metric	Description	-----	Win Rate	% of trades with positive P&L	Profit Factor	Gross profits
	/ Gross losses	Total Return	(Final capital - Initial) / Initial	Max Drawdown	Largest peak-to-trough decline	
Sharpe Ratio	Risk-adjusted return (annualized)	Avg Win / Avg Loss	Mean P&L of winning vs losing trades			

6.3 Stop-Loss Comparison

Two strategies were compared:

- 1 **Hold strategy:** Hold for full 10 days regardless of intraday P&L

- 2 **Stop-loss strategy:** Close at +10% profit or -5% loss

The hold strategy consistently outperformed the stop-loss strategy in backtesting, as stop-losses frequently triggered during normal intraday volatility before the anticipated move occurred. The hold strategy was therefore selected for production.

Note: For live trading, a +10% take-profit auto-close was added as a practical measure to lock in gains, while no stop-loss is enforced on the downside.

7. Signal Generation & Trading Logic

7.1 Core Signal: IV-RV Spread

The fundamental insight is comparing two volatility estimates:

- 1 **GARCH Realized Volatility (RV):** What the model predicts actual price movements will be
- 2 **Market Implied Volatility (IV):** What the options market is pricing in

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Spread = GARCH_RV - Market_IV
If Spread > 0: Options are UNDERPRICED · BUY volatility (long straddle)
If Spread < 0: Options are OVERPRICED · SELL volatility
```

7.2 Why Long Straddles?

A long straddle profits when the underlying stock moves significantly in **either direction**. It loses when the stock stays flat (due to time decay / theta).

$$\text{Straddle P\&L} = |\text{Stock Move}| - \text{Premium Paid} - \text{Theta Decay}$$

When GARCH predicts higher realized vol than the market expects:

- The stock is likely to move more than priced in
- The straddle premium is "cheap" relative to expected movement
- Expected value is positive

7.3 Signal Strength

Signal strength is computed as the **relative spread**:

$$\text{Strength} = |\text{Spread}| / \max(\text{Market_IV}, 0.01)$$

Signals are ranked by spread (strongest positive spread = best buy-vol opportunity).

8. Live Scanning & Execution

8.1 Scanner Pipeline

Every scan cycle:

- 1 Iterate through all 42 tickers
- 2 For each ticker: a. Fetch current options chain from Yahoo Finance b. Find nearest expiration ~14

- days out c. Try ATM and ± 1 strikes d. Compute straddle cost (call + put premium $\times 100 + \$1.30$ commission) e. Filter: must fit within \$150 budget f. Fit GARCH model on full price history g. Compute spread = GARCH conditional vol - 21-day rolling realized vol
- 3 Sort all opportunities by spread (highest first)
 - 4 Return top 8

8.2 Execution via Webull

Orders are placed through the Webull OpenAPI:

- **Authentication:** HMAC-SHA1 signature on every request
- **2FA:** Token-based verification via Webull mobile app
- **Order type:** LIMIT orders at last traded price
- **Legs:** Two separate SINGLE-leg orders (call + put)
- **Time-in-force:** GTC for buys, DAY for sells

8.3 Auto-Close Monitor

A background daemon (`broker/auto_close.py`) runs continuously:

- **Market hours (9:30 AM - 4:00 PM ET):** Poll every 30 seconds
 - **Off-hours:** Poll every 5 minutes
 - **Trigger:** If any straddle's total unrealized P&L $\cdot +10\%$, auto-sell both legs
 - **Logging:** All checks logged to `cache/auto_close.log`
 - **Trade records:** Closed trades saved to `cache/closed_trades.json`
-

9. Risk Management

9.1 Position Sizing

- Maximum 90% of capital deployed per trade
- Budget capped at \$150 per trade
- Commission-aware sizing (\$0.65/contract/leg)

9.2 Diversification

The scanner ranks 42 tickers but executes only the **single best signal**. This concentrates risk but maximizes edge on the highest-conviction opportunity.

9.3 Time-Based Risk

- 10-day maximum holding period limits exposure to prolonged theta decay
- Options selected with ~ 14 -18 days to expiration, avoiding rapid time decay near expiry

9.4 Known Risks

| Risk | Mitigation | |-----|-----| | Theta decay | GARCH spread must exceed theta cost | | Liquidity | Scanner filters for minimum volume | | Model risk | GJR-GARCH captures asymmetry; Student-t captures fat tails | | Execution risk | Limit orders at last price; GTC for fills | | Data risk | 12-hour cache freshness; fallback to stored data |

10. Performance Summary

10.1 Model Accuracy

- GARCH conditional volatility correlates 0.5-0.8 with realized volatility on validation sets across tickers
- GJR-GARCH selected over symmetric GARCH ~70% of the time, confirming leverage effects

10.2 Strategy Design Decisions

Decision	Chosen	Rejected	Reason	----- ----- ----- -----	Holding period	10 days	5 days, weekly
	Better capture of vol events		Exit strategy	Hold to expiry	Stop-loss (10%/5%)	Stop-loss underperformed	
Model	GARCH-only	GARCH + Transformer	GARCH-only more robust		Order type	Two single legs	
	Multi-leg combo	Webull API compatibility					

11. Appendix: Mathematical Reference

A. Log Returns

$$r_t = \ln(P_t / P_{t-1})$$

B. GARCH(1,1) Variance Equation

$$\sigma_t^2 = \omega + \alpha \sigma_{t-1}^2 + \beta \sigma_{t-1}^2$$

C. GJR-GARCH(1,1,1) Variance Equation

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \sigma_{t-1}^2 + \beta \sigma_{t-1}^2$$
$$I_{t-1} = 1 \text{ if } r_{t-1} < 0, \text{ else } 0$$

D. Annualization

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

E. Black-Scholes Straddle Price

$$\text{Straddle} = \text{Call}(S, K, T, r, \sigma) + \text{Put}(S, K, T, r, \sigma)$$

F. Student-t Log-Likelihood

$$l(\sigma, \sigma^2) = \frac{1}{n} [\ln \sigma \cdot ((n+1)/2) - \ln \sigma \cdot (n/2) - \frac{1}{2} \ln(\sigma \cdot (\sigma^2 - 2) \cdot \sigma_t^2) - ((n+1)/2) \cdot \ln(1 + \sigma_t^2 / ((\sigma^2 - 2) \cdot \sigma_t^2))]$$

G. AIC Model Selection

$$\text{AIC} = 2k - 2 \cdot \ln(L)$$

Lower AIC = better model (balances fit vs complexity).

Document generated from source code analysis of the volatility trading system. All model parameters and logic correspond to the production codebase as of April 27, 2026.